KNECK OFFICIAL

Woodwork — Paper 1

444/1 • ORIGINAL REVISION SET 01 • 2 hours 30 minutes • 100 marks

Name: _____ Class: _____ Date: _____

Independent KCSE-style practice. Not issued, endorsed or predicted by KNEC. These are original questions, not leaked or past official papers. Component numbering is checked against the KNEC 2026 timetable. A teacher should review these practice materials before formal classroom assessment.

CANDIDATE INSTRUCTIONS

Answer all questions in Sections A and B and ANY TWO in Section C. Use labelled freehand sketches where asked. No powered tool use is needed for this theory paper.

Use a separate answer booklet. Show working where appropriate. The marking guidance begins after the END OF QUESTIONS page; do not consult it until you finish. Practical and project components require the stated equipment and teacher supervision, not merely a printed worksheet.

SECTION A

40 marks; answer all.

1. Name two hardwoods and two softwoods commonly used in joinery, distinguishing botanical grouping from literal hardness. [5 marks]

2. State five reasons for seasoning timber. [5 marks]

3. Describe five workshop safety rules. [5 marks]

4. State five common timber defects. [5 marks]

5. Name five marking-out tools and their uses. [5 marks]

6. Explain the difference between a rip saw and a crosscut saw and give three safe sawing practices. [5 marks]

7. State five factors affecting choice of adhesive. [5 marks]

8. Give five ways to reduce wood waste in a school workshop. [5 marks]

SECTION B

30 marks; all questions.

9. Sketch a through mortise-and-tenon joint. Label shoulder, tenon, mortise and cheek. Describe its marking, cutting and trial assembly. [15 marks]

10. A shelf needs two boards 600×200×20mm and three supports 200×40×20mm. (a) Produce a cutting list. (5) (b) Calculate net timber volume in m³. (5) (c) Calculate volume including 15% waste. (3) (d) Give two reasons for allowing waste. (2) [15 marks]

SECTION C

Choose two; 15 marks each.

11. Describe preparation and application of a clear water-based finish on a wooden shelf, including safety and quality checks. [15 marks]

12. Compare air seasoning and kiln seasoning; explain suitable storage of seasoned timber. [15 marks]

13. Describe design and construction stages for a small wall shelf, including working drawing, materials, joints, assembly and evaluation. [15 marks]

END OF QUESTIONS

MARKING GUIDANCE

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Suggested answers and assessment guidance, not an official marking scheme. Accept equivalent correct reasoning, valid examples and properly justified alternative methods. Marks for an item must not exceed its printed allocation. For essays use relevant textual evidence and the rubric; no plot summary alone.

SECTION A

1. (5 marks) Hardwoods e.g. mahogany/teak/mvule; softwoods pine/cypress/cedar two each4; botanical classification not a guarantee of physical hardness1.

2. (5 marks) Dimensional stability,reduce weight,improve strength/workability,reduce decay risk,improve glue/finish acceptance; five.

3. (5 marks) Eye protection,secure work,guards,suitable clothing,no untrained machine use,clear workspace,dust control; five.

4. (5 marks) Knots,shakes,checks,splits,warping,cup,bow,twist,decay; five.

5. (5 marks) Rule measure,try square right angles,marking gauge parallel line,mortise gauge mortise widths,knife precise line; five paired.

6. (5 marks) Rip along grain/crosscut across grain with appropriate tooth action (2); secure work,hand clearance,support offcut,controlled stroke,three (3).

7. (5 marks) Materials,moisture exposure,strength,open time,clamping,cure conditions,safety; five.

8. (5 marks) Cutting list,nesting/marking,accurate measurements,reuse offcuts,repair,recycle dust safely,maintain sharp tools; five.

SECTION B

9. (15 marks) Recognisable proportioned sketch3; labels4; matching gauge/reference faces2; accurate mortise cutting2; tenon sawing2; trial fit/square check2.

10. (15 marks) List correct quantities/dimensions/units5. Boards2×.6×.2×.02=.0048; supports3×.2×.04×.02=.00048; total.00528$m^3$5. ×1.15=.006072$m^3$3. Saw kerf/defects/planing allowance,two2.

SECTION C

11. (15 marks) Prepare/repair surface3; progressive sanding with grain/dust removal3; test/appropriate thin coats and drying3; de-nib/recoat as specified2; ventilation/PPE/waste safety2; inspect smooth/even finish2.

12. (15 marks) Air stacking/stickers/ventilation3; kiln controlled heat/humidity3; time/control/cost/defect comparison four points4; dry supported level protected ventilated storage and stock rotation5.

13. (15 marks) Brief/measure/user needs2; dimensioned drawing/cutting list3; suitable materials/joints3; safe accurate making3; square assembly/finish2; load/quality evaluation2.

REVISION CHECK

Mark your work, list three misconceptions, revise those topics and attempt the incorrect questions again without the guide. These topics are revision priorities, not a forecast of the real examination.

Paper/component reference: https://www.knec.ac.ke/wp-content/uploads/2026/02/2026-KCSE-Timetable.pdf